

Computer Programming

Lab Report 09

Lab title

Introduction to Functions in C++

1. Objective

The objectives of this lab are:

- To understand the concept of functions in C++
- To learn function prototypes
- To understand function calling and function definition
- To use functions with and without parameters
- To understand return values in functions

2. Introduction

- Functions are an important feature of C++ programming that divide a large program into smaller, reusable modules. Each function performs a specific task and can be called whenever required.

Functions help in:

- Improving code reusability
- Making programs easier to understand
- Reducing code repetition
- Simplifying debugging

A function consists of:

- Function name
- Return type
- Parameters (optional)
- Function body

The `main()` function is the starting point of every C++ program.

3. Function Prototype

A function prototype tells the compiler about:

- Function name
- Return type
- Number and type of parameters

Syntax:

```
returnType functionName(parameterList);
```

Examples:

```
int main();  
float divide(int a, int b);  
void square(float x);
```

4. Program 1: Function Without Parameters

```
#include <iostream>  
using namespace std;  
  
void asterisk();  
  
int main()  
{  
    asterisk();  
    return 0;  
}  
  
void asterisk()  
{  
    for(int i = 0; i < 10; i++)  
    {  
        cout << "*";  
    }  
}
```

Output:

5. Program 2: Function with One Parameter

```
#include <iostream>  
using namespace std;  
  
void asterisk(int n);  
int main()  
{  
    asterisk(12);  
    return 0;  
}
```

```

void asterisk(int num)
{
    for(int i = 0; i < num; i++)
    {
        cout << "*";
    }
}

```

Output:

6. Program 3: Function with Two Parameters

```

#include <iostream>
using namespace std;

int add(int num1, int num2);

int main()
{
    int result;
    result = add(3, 5);

    cout << result;

    return 0;
}

int add(int num1, int num2)
{
    return num1 + num2;
}

```

Output:

8

7. Result

- All programs were successfully compiled and executed. Correct outputs were obtained in each case.

Task 1: Function Introduction and Declaration

Problem Statement

- Declare a function called greet() that takes no parameters and returns nothing. Display a greeting message inside the function and call it from main().

Program

```

#include <iostream>    // Header file for input and output
using namespace std;  // Use standard namespace

void greet();        // Function prototype

int main()
{
    greet();          // Function call

    return 0;         // End program
}

// Function definition
void greet()
{
    cout << "Welcome to Computer Programming Lab!" << endl;    //
    Display greeting message
}

```

Output

Welcome to Computer Programming Lab!

Task 2: Basic Function Declaration and Call

Problem Statement

- Declare a function called sayHello() that takes no parameters and returns nothing. Display the message "Hello, World!".

Program

```

#include <iostream>    // Header file for input and output
using namespace std;  // Use standard namespace

void sayHello();      // Function prototype

int main()
{
    sayHello();        // Function call

    return 0;         // End program
}

// Function definition
void sayHello()
{
    cout << "Hello, World!" << endl;    // Display message
}

```

Output

Hello, World!

Task 3: Sum and Average Using Function

Problem Statement

Write a C++ program that:

- Takes 10 integer values from the user
- Stores values in a 1D array
- Uses a function calculateSum()
- Returns the sum of array elements
- Calculates and displays average in main()

Diagram

```
Index:  0  1  2  3  4  5  6  7  8  9
      +---+---+---+---+---+---+---+---+---+
Array = | 12 | 15 | 10 | 20 | 18 | 25 | 30 | 14 | 16 | 20 |
      +---+---+---+---+---+---+---+---+---+
```

Sum = 180
Average = 18

Program

```
#include <iostream>    // Header file for input and output
using namespace std;  // Use standard namespace

// Function prototype
int calculateSum(int arr[], int size);

int main()
{
    int numbers[10];    // Declare array
    int sum;            // Variable for sum
    float average;      // Variable for average

    // Input array elements
    cout << "Enter 10 numbers:" << endl;

    for (int i = 0; i < 10; i++)    // Loop for input
    {
        cin >> numbers[i];    // Store numbers in array
    }

    // Function call
```

```

    sum = calculateSum(numbers, 10);

    // Calculate average
    average = sum / 10.0;

    // Display sum
    cout << "Sum = " << sum << endl;

    // Display average
    cout << "Average = " << average << endl;

    return 0;    // End program
}

// Function definition
int calculateSum(int arr[], int size)
{
    int total = 0;    // Variable for total sum

    for (int i = 0; i < size; i++)    // Loop through array
    {
        total = total + arr[i];    // Add array elements
    }

    return total;    // Return total sum
}

```

Sample Output

Enter 10 numbers:

12
15
10
20
18
25
30
14
16
20

Sum = 180

Average = 18

8. Conclusion

In this lab, we learned:

- The basic concept of functions in C++
- Function prototypes and their importance
- Function calling and definition
- Functions with zero, one, and multiple parameters
- Functions returning values

